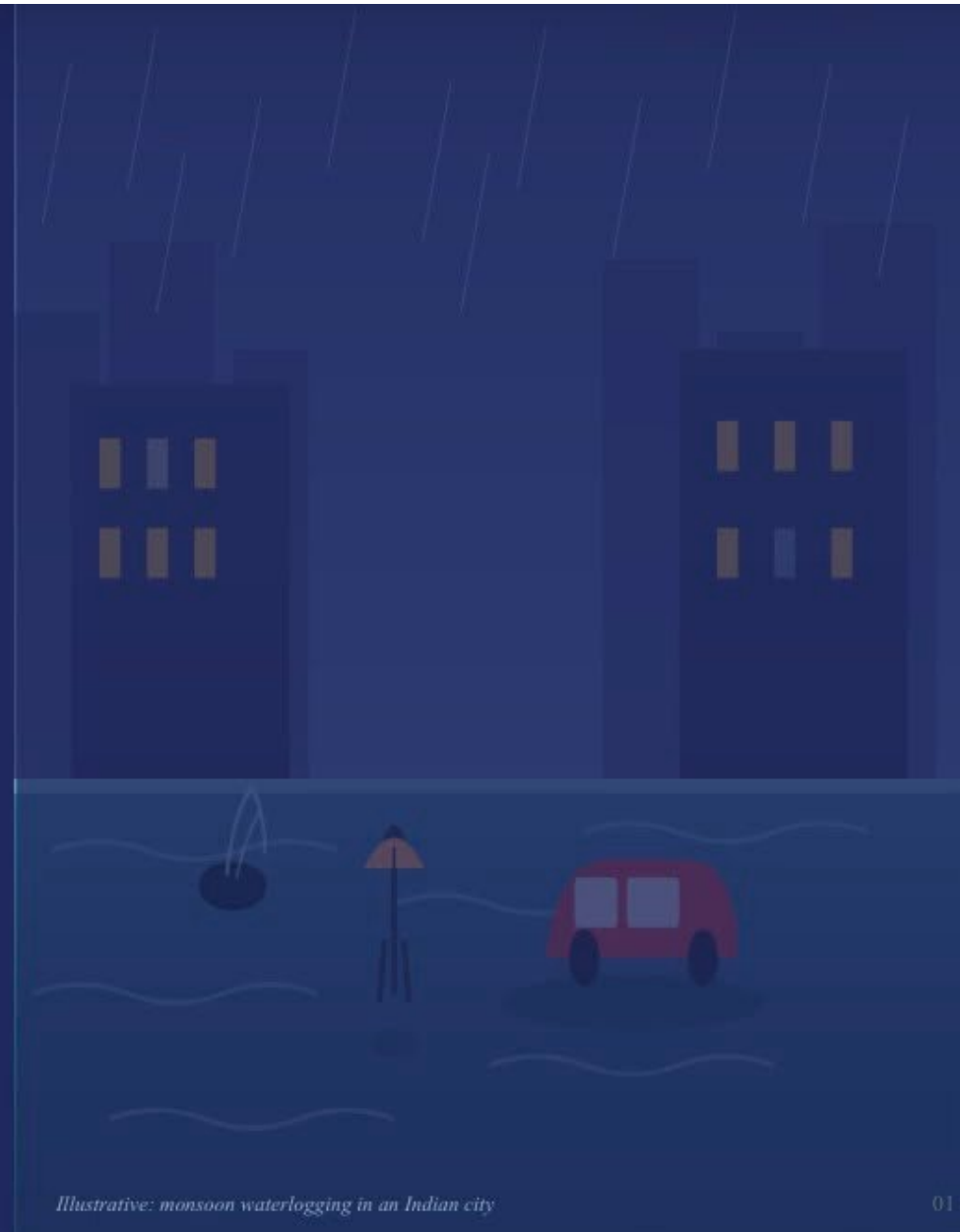


Urban Flood Nowcasting System

Coupling Rainfall Nowcasts with Urban Drainage Networks for Street-Level Flood Prediction

Theme: Disaster Management



Illustrative: monsoon waterlogging in an Indian city

BACKGROUND

Rain isn't the only variable — drainage is

Indian cities repeatedly flood not just from extreme rainfall, but from silted drains, undersized pipes, and surcharged outfalls. Existing systems forecast rainfall in isolation — they don't tell authorities which streets will actually go under water, or when.

944 mm

rainfall in a single day —
Mumbai, 26 July 2005

0–3 hrs

realistic nowcast lead time for
actionable warnings

1000s

of drain segments a city model
must account for



Coimbatore, monsoon waterlogging



The gap today

- Rainfall forecasts exist — flood forecasts at street level don't
- Drainage capacity data sits unused in silos
- No fused, real-time view for disaster managers
- Warnings arrive too late for evacuation or traffic re-routing

This is the everyday reality nowcasting has to address — not an abstract rainfall total, but a specific street, at a specific time, under a specific depth of water.

OBJECTIVE

What the system must deliver



Forecast rainfall

0–3 hr nowcasts at fine spatial resolution using radar/satellite + gauge data.



Model the drains

Couple forecasts with real pipe capacity, gradients & outfall points.



Localise the risk

Predict flooding at street / ward level — not just a city-wide rainfall number.



Warn in time

Give disaster managers and citizens enough lead time to act.

ARCHITECTURE

End-to-end system pipeline



Update cycle: rolling 10–20 min refresh as new radar/gauge data arrives

COMPONENT 01

Rainfall Nowcasting Layer



Inputs

- IMD Doppler weather radar reflectivity
- INSAT satellite rainfall estimates
- Automatic weather station (AWS) & rain-gauge network
- Historical rainfall archives for training

Model approach

ConvLSTM / optical-flow extrapolation over radar image sequences — captures both spatial rainfall structure and its motion. Indian studies show ~75% correlation skill at 20 min lead time, ~35% retained at 100 min — enough for a 0–3 hr nowcast window.



Output

Gridded rainfall-intensity forecast covering the city, refreshed every 10–20 minutes, feeding directly into the drainage coupling layer.

0–3 hr

lead time

10–20 min

refresh cycle

75%

skill @ 20 min

COMPONENT 02

Drainage & Hydraulic Coupling Layer

*This is what separates a flood *forecast* from a rainfall forecast: rain is routed through the city's actual drainage network to see where it backs up.*



1D Pipe Network Model

- Pipe diameters & gradients
- Junction & manhole locations
- Outfall / discharge points



2D Surface Flow

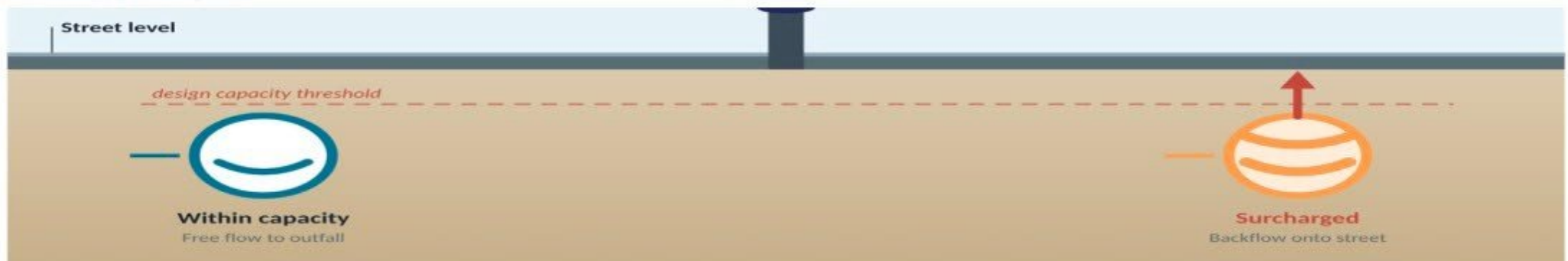
- DEM-based overland routing
- Land-use & imperviousness
- Fast surrogate vs full SWE solver



Coupled Inference

- Capacity-threshold surcharge flags
- Street-level depth estimate
- Rolling re-computation on new rain

REFERENCE DIAGRAM



TECH STACK

Suggested technology stack



Data & Storage

PostgreSQL/PostGIS, GeoTIFF/NetCDF for grids, object storage for radar frames



Modelling

Python, PyTorch/TensorFlow (ConvLSTM), scikit-learn, PySWMM / SWMM-style drain logic



Geospatial

QGIS, GDAL, Leaflet/Mapbox for map rendering, OpenStreetMap road network



Dashboard & Alerts

React front-end, FastAPI/Node backend, WebSocket push, SMS/Push via Twilio-style gateway

DATA SOURCES

Where the data comes from



IMD Radar & AWS Network

Doppler radar reflectivity and real-time rain-gauge readings (e.g. mumbairain.tropmet.res.in).



Municipal Drain GIS

Pipe layouts, diameters, gradients from city corporation portals — availability varies by city.



Historical Flood Records

Past waterlogging incident reports for model validation and threshold calibration.



OpenStreetMap

Road network and elevation context where municipal DEM data isn't public.

BUILD PLAN

What's realistic in a phased build

01

Idea

Pick one flood-prone ward with public drain/GIS data (e.g. Mumbai's Hindmata-Dadar)

02

Prototype

Lightweight nowcast (persistence/extrapolation) + rule-based drain capacity model

03

Integration

Fuse both layers; render predicted flooding on a street-level map

04

Polish

Dashboard UX, alert thresholds, ward-wise summary for judges' demo

Full CFD-grade hydraulics and a full city rollout are longer-term roadmap items — not part of the initial build.

EXPECTED OUTCOMES

What success looks like



Street-level lead time

1–3 hr advance warning per street/ward, not just a city-wide rainfall number



Actionable dashboard

Live view for municipal & disaster management authorities to prioritise response



Public alerts

SMS/app notifications to residents and commuters in at-risk zones



Scalable framework

Same nowcast + drainage coupling approach portable to other flood-prone cities



Thank You